

Sensemaking: What is it today?

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ABSTRACT

Sensemaking is a common activity in the analysis of a large or complex amount of information. It has also been an active area of research for at least 25 years. Such an active area of HCI research over a quarter century asks fundamental questions about how do people come to understand difficult sets of information? How do they find the data? How do their sensemaking frames help (or hinder) their ability to understand? The information workplace is increasing dominated by high velocity, high volume, complex information streams. At the same time, understanding how sensemaking operates has become an urgent need in an era of increasingly unreliable news and information sources. While there has been a huge amount of work in this space, the research involved is scattered over a number of different domains with differing approaches. This workshop will focus on the most recent work in sensemaking, the activities, technologies and behaviors that people do when making sense of their complex information spaces. We will also attempt to synthesize sensemaking work over the past several years. In the second part of the workshop we will synthesize a cross-disciplinary view of how sensemaking works in people, along with the human behaviors, biases, proclivities, and technologies required to support it.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**.

KEYWORDS

sensemaking, generative ai

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1 BACKGROUND

Making sense of the world is a very common activity. It happens whenever you confront a new, complex problem, or a conflicting set of diverse information sources. At work, your boss asks, “Can you give a presentation next week on how wireless will affect our business?” Or perhaps you join a new committee and wonder “Who are these people?” “Who is in charge?” “What is our mission?” Perhaps you move to a new neighborhood, and try to make sense of the streets, schools, parks, shopping, and neighbors. Likewise, the imperative for citizens to make sense of competing and conflicting news sources is important. Each of these is a sensemaking task for a particular domain.

The rise of AI, chatbots, and personal assistants have all focused on providing simple informational results, rather than helping with the hard work of connecting the dots and accomplishing complex tasks. The availability of data has continued to increase exponentially, but has led to few transformative novel paradigms for making sense of it.

More recently, sensemaking has become an important factor in our ability to read and understand world events. Consider the spread of fake news as a kind of challenge to our ability to make sense of competing (and often contradictory) information. We see a fragmenting of attention, a lack of trust, and an inability of people to comprehend scientific information. In a world where there is a new rise of big data, the need for sensemaking is greater than ever.

On the other hand, the tools that we have for making sense of information are no better—and in many cases are now worse—that they have ever been. For example, this past year Large Language Models (e.g., ChatGPT, Bard, or Claude) seem to offer entirely new capabilities for making sense of complexity through their new summarization and abstraction capabilities. Yet these new tools pose new challenges for sensemaking because of their lack of true understanding of the text they generate, and the apparent high quality of what they write. The rise of AI, chatbots, and personal assistants have all focused on providing simple informational results, rather than helping with the hard work of connecting the dots and accomplishing complex tasks. The availability of data has continued to increase exponentially, but has led to few transformative and novel paradigms for making sense of it.

These kinds of information sensemaking tasks presage a process of collecting, understanding, and organizing data. Often the information can be organized into a fairly simple structure, one that helps to solve the problem. But as we well know, the process is

sometimes ill-defined, iterative and complex: information retrieval, organization and task redefinition all interact in sometimes subtle ways. All of these behaviors lead to the creation of sense; that is, the process of sensemaking.

Sensemaking can be a core professional task in itself, as it is for researchers, designers, or intelligence analysts. It arises when we change our place in the world or when the world changes around us. It arises when new problems, opportunities, or tasks present themselves, or when old ones resurface. It involves finding the important structure in a seemingly unstructured situation. It is an activity with cognitive and social dimensions, and has informational, communicational, and computational aspects.

Such an important activity deserves any support it can get from information technologies. Indeed, many major thrusts of HCI can be seen as contributing salient perspectives and technologies. Visualization aims to help people see structure in data or information; information retrieval is needed to find. Combining all of these aspects of information use defines the behaviors of sensemaking—what people do to make sense of the information in their world.

For example, a seminal early hypermedia work of Notecards [7, 13] was aimed at helping work in design, research and intelligence analysis. Notecards provided tools for creating structures from initial data, then evolving those representational structures to better fit the task. Search tools and visualizations made examining the information space straightforward. In essence, Notecards was an early sensemaking system reflecting each of the key aspects of sensemaking behaviors—access to a relatively large amount of information, tools to structure and organize that information, visualizations to help see the structure of the content, and information retrieval mechanisms to provide search support.

Another key role that tools and visualizations can play in support of collaborative sensemaking, is to help attenuate known cognitive biases and limitations of distributed teams making decisions. For example, an experiment on collaborative intelligence analysis via the CACHE system - a proof of concept prototype - showed that known biases such as anchoring and confirmation bias can be tamed via purposely designed collaborative visualizations that reinforce the visibility of nonconfirming evidence across remote intelligence analysts working on the same case [2]. Also in the context of distributed collaboration, but in the domain of emergency management planning, other research has shown that collaborative visualizations can be engineered to enhance the sharing and awareness of both content and process knowledge in groups [4]. The research shows that the improvements in performance of distributed teams can be improved, in some respects, even beyond the performance of comparable face-to-face groups.

2 SENSEMAKING TODAY

There has been a recent increase of focused interest in sensemaking, spurred by at least two forces. One has been the push of the information explosion from the WWW. As is evident, web-based tools have radically changed end-user ability to access large amounts of information; this in turn fundamentally changes the ways in which people seeking to understand complex problems think about the issues. Consequently, both the Library and Information Science

(LIS) and HCI communities have begun converging on projects to help people make sense of the information resources now available.

The other force has been a pull from the post-9/11 intelligence community, as the need to make sense of terrorism related intelligence data has become internationally urgent.

As we have begun to realize, sensemaking extends far beyond individuals making sense of their information spaces. Groups also frequently need to work together to understand larger issues, combining information from many information sources, and bringing together a synthetic understanding that melds different points of view. Group problem solving work in CSCW often involves support for sensemaking activities, and there are active research efforts in understanding what sensemaking groups currently do, and what tools might support them. [1, 6]

What's more, as data is increasingly harvested from multiple sources on an ad-hoc basis, the organization and interpretation of the data now needs to account for the quality of the data from each of the sources. When the quality or (or noise in) the data varies, it is possible to create helpful sensemaking tools that indicate data quality? [10]

Data sensemakers often also find having to account for the provenance of data. That is, a representation of the data lineage to show what evidence is coming from what source, especially when the bias-per-source is expected to vary. [8]

In this workshop we will be asking about the current state of sensemaking with respect to the skills required and the tools needed to support sensemaking. We will consider:

1. Tools to retrieve information: e.g., find a how-to video or recipe for a cake in Youtube; discover the facts behind a political news via Twitter or search engines.
2. Tools for visualizing the information space: Such as tools for seeing the shifts of voter groups toward left or right over the years. For instance, could we easily find tools to transform data into stacked histograms, or whatever visualization is most appropriate.
3. Tools for individual decision-making: What tools and procedures can we use to evaluate alternative product to buy, alternative stocks to invest in, etc.
4. Tools for collaborative decision-making: Can we decide what vacation to have with friends, discover potential terrorists with other intelligent analysts, etc. While collaborative tools have great promise, the state of the art seems as yet underserved by our research.

3 WEBSITE

<https://sites.google.com/view/chi2024-sensemaking-workshop/home> will be the go to site for access to the papers, people, and schedules of the workshop.

4 WORKSHOP GOALS

The workshop will include research in areas such as:

1. how do people make sense of complex sets of information? (behavior studies and tool use)
2. issues of representation creation, evolution and use over time
3. implicit and explicit aspects of sensemaking

4. group sensemaking: including different levels of social aggregation, from individual, to group, to large social contexts - both static and evolving problem environments

5. how sensemaking fits into other knowledge work (information gathering, decision making)

6. what is sensemaking today? (In particular, what other sense-making schools of thought are there, and how can we mutually inform each others work?)

From this meeting of the minds, the Sensemaking workshop has several desired outcomes: First, we will create working relationships between researchers whose work focuses on aspects of sensemaking. While we certainly hope to bring together those working within the HCI community, we would like to try to bring in some researchers from other disciplines as well, including Library and Information Science (LIS) [3] and Organizational Theory [14] and psychology (e.g., cognitive/problem solving research).

Our second outcome is to enrich our understanding of sense-making activities. This includes striving for a shared understanding of the different notions of sensemaking, laying out and structuring the space of varieties of sensemaking (e.g., different levels of social aggregation, static vs. dynamic contexts), articulating their commonalities and differences.

And our third goal is to draw from this is a greater understanding of design implications for improved sensemaking tools, systems and designs. There is a clearly emerging demand for tools for verifiability and trustability of facts shared on public media channels. For example, a new generation of tools is emerging to allow journalists to spot inaccurate or fake news by leveraging ML algorithms and visualizations. Can we take advantage of these tools in our everyday sensemaking tasks as well.

5 WORKSHOP PLANS

This year we plan to hold a 1-day workshop on sensemaking, following on the tradition of previous CHI workshops on this topic (2005, 2007, 2009). Before the workshop, we'll be collecting the position papers to be presented at CHI, starting up pre-workshop email discussions, bringing all participants to a common ground of understanding what varieties of sensemaking will be presented. The workshop itself will be an exercise in group sensemaking in real-time (a common application of sensemaking practice). Based on the pre-workshop papers, submissions, and research activities, we will break up into teams to find the central organizing theme and structure of the field. In the afternoon, we will synthesize the findings of each team and create a larger, more diverse and explanatory document that tells us how and where sensemaking research is being done.

6 WORKSHOP STRUCTURE

The workshop will be organized in a hybrid fashion with participants having an option between an in-person participation or via a Zoom/alternative similar platform. Since the workshop has multiple activities, we will consider using different tools like Google Workspace (slides, docs etc.), Figma, Miro, breakout rooms, etc. to enable closer collaboration. Most of the content (position papers, collaborative activities, videos of discussions) will be made available

to participants, pending institutional approvals, for asynchronous consumption.

The workshop is intended as a one day workshop with pre and post workshop plans as discussed in the previous and next section, respectively. The timeline of the day is can be seen in Table 1.

At the day of workshop, participants will begin by introducing themselves, their work, and their hopes/dreams from this workshop. Next, we will encourage the participants to explore and pose open research questions aligned with the workshop goals. These may be beyond their submitted position papers. We will collectively discuss and dot vote the ones we would like to pursue during the workshop.

Participants will next be encouraged to re-group into smaller groups to focus on one of the research questions and brainstorming "how might we" solve for that research question. To aid brainstorming, participants would be provided with dimensions: literature review (theoretical approaches and knowledge, existing interventions); and solutions (future research, and interventions). We plan to use flipboards and one of the hybrid solutions (slides, miro etc.) for collaborative work. At the end of this brainstorming, the groups will share their outcomes to the workshop attendees.

Post shareout, the groups will be encouraged to cross-pollinate and discover opportunities to collaborate with each other across the different groups, leading to final shareout. The workshop will end with discussion about where to go next ? This may include posters, workshops, articles, and other avenues.

7 POST-WORKSHOP PLANS

After all of the accepted workshop papers are presented, we plan on doing a sensemaking session about what was just presented—in the vein of “making sense of sensemaking.” The immediate output from this will be a poster to be presented at CHI 2025. We'll discuss other possible venues for publishing this collection of papers (as a special journal issue, a collected volume, or as an e-book for future sensemakers to find and use easily).

8 ACCESSIBILITY

We will encourage the authors to submit an accessible version of their position papers. We will use a platform that enables captions for the videos, and potentially during other collaborative parts of

Time	Activity
9:00-9:30	Setting the stage
9:30-10:00	Introductions
10:00-10:20	Break
10:20-11:00	Gather open research questions, and discuss
11:00-11:30	Dot-vote, prioritize, and re-group
11:30-12:00	Literature review
12:00-12:30	Lunch break
12:30-13:00	How Might We - Pursuing future next steps and share-out
13:00-13:30	Coffee Break
13:30-14:00	How Might We - Cross-pollinate and share-out
14:00-14:30	Next steps and fin!

Table 1: Workshop schedule

the program too. In case of any further accessibility needs, we will engage with the Accessibility Chair to enable a positive experience.

9 CALL FOR PARTICIPATION

We are seeking workshop papers on topics in the area of sensemaking for a workshop to be held at CHI 2024 in Hawai'i, USA. Making sense of information is central to HCI as people look to understand complex systems, domains and problems. Broadly, we take the topic of sensemaking to mean understanding of how people collect and organize information for analysis and synthesis, and the tools and processes they follow when doing this.

Thus, sensemaking, per se, is everywhere in the systems we build and in the domains we study. Whenever people need to function well with data, making sense of the information is often a central task. We are interested in both individual and group sensemaking practices—from how one person figures out where to go when visiting Oahu, or what has happened recently on Maui, up to large, collaborative group sensemaking where teams of people assemble and interpret large, complex, interlocking sets of data. Representative tasks include the practice of people who deal with sensemaking hand-offs (e.g., in a medical setting) or in analytical areas (e.g., making sense of financial data for forensic purposes).

In particular, what are the tools, techniques and best practices of people who need to make sense of a large amount of complex information? What issues of scale, complexity and coordination arise that are particular to making sense of a complex world?

10 ORGANIZERS

Daniel M. Russell has been working in the area of sensemaking since the early 1990s. His publication of *The cost structure of sensemaking* [12] in 1993 led to a stream of research in this area. Now a Senior Research Scientist at Google, he primarily studies how people formulate information needs and satisfy them with online research tools and databases. He has run three earlier CHI workshops on sensemaking, one of which led to a special edition publication of the journal *Computer-Human Interaction* [11].

Niki Kittur is an Associate Professor and Cooper-Siegel Chair in the Human-Computer Interaction Institute at Carnegie Mellon University. His research explores a future that scales sensemaking beyond the limits of a single individual's mind by: 1) distributing sensemaking among many people and machines; 2) enabling people to build on the sensemaking that others have already done; and 3) seamlessly integrating human and machine cognition to make sense of large information spaces. He is also a co-founder of DataSquid, a startup that supports sensemaking by bringing the power of intuitive touch and physics to data visualization.

Nitesh Goyal (Tesh) leads research at the intersection of HCI and AI, in the Responsible AI team of Google Research. His work at Google has led to launch of ML based tools like AI Test Kitchen as a sandbox to upload generative AI applications, MakerSuite as a tool for developers to leverage Google's generative AI capabilities without coding, Harassment Manager to empower targets of online harassment, and ML based moderation to reduce online toxic content production on platforms like OpenWeb. He also introduced design metaphor of "Sensemaking Translucence" [4] to reduce biased sensemaking in criminal justice while managing interruptions

appropriately [5]. His research has been supported by German Govt. Fellowship, National Science Foundation, and MacArthur Genius Grant.

Laura Koesten is a Postdoc at the University of Vienna (Austria) in the Research Group for Visualization and Data Analysis (Faculty of Computer Science) and an affiliate researcher at King's College London, UK. Her research explores the characteristics of sensemaking with data in different contexts, such as data discovery, data reuse, and data visualizations. She uses verbal and written summarization techniques to surface sensemaking activities and studies subjective experiences of sensemaking in interactions with datasets and data visualizations. She is PI of the Talking Charts project, which investigates how charts related to climate change and COVID-19 are created and understood by public as well as by scientific audiences. [9]

Dr. Pengyi Zhang is an associate professor at Department of Information Management, Peking University. She holds a Ph.D. and a MLS from College of Information Studies, University of Maryland. Her research expertise include information and knowledge organization, information/data seeking and sensemaking, and user-centered design and evaluation. She is particularly interested in how people seek for and make sense of data and information. Her recent research investigates the how individuals in groups and communities make sense together to negotiate meaning, deal with conflicts, and form consensus. She has received grants from National Science Foundation of China, Chinese Library Association, and OCLC/ALISE Library and Information Science Research Grant.[15]

Dr. Michael Xieyang Liu is a research scientist at Google in the Pittsburgh office. He recently obtained his PhD from Carnegie-Mellon University with his thesis *Tool Support for Knowledge Foraging, Structuring, and Transfer During Online Sensemaking*. His work has integrated several different sensemaking tools and systems into a coherent whole, establishing patterns for bridging the gap between rapidly evolving mental models in peoples' heads and the externalization of those models. Specifically, he designs and builds interactive systems to reduce the costs and increase the benefits of externalization, thereby capturing more of the cognitive work that users engage in while making sense of information.

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